Distributed Event Reasoning and Information Fusion for Decision Support

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Command, Control, Communications, Computers, Cyber, Intelligence, Surveillance, and Reconnaissance (C5ISR) and Naval Space — Intelligence, Decision Superiority, Command and Control (C2), and Combat Systems

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Problem. Command staff may receive the same event reports in different orders, with duplicate, delayed, or missing messages. One afloat node receives a report of an unidentified vessel near an exercise area, followed by a message identifying it as scheduled support traffic; a shore node receives those reports in reverse order. Conventional retrieval-augmented assistants process whatever each node has seen, so they can build inconsistent narratives even after the same reports eventually arrive. Event extraction from a centralized collection does not solve the additional problem of synchronizing evolving text-derived knowledge across disconnected nodes.

Central hypothesis. Representing extracted events as mergeable updates with explicit uncertainty and source identity will make distributed information fusion less sensitive to arrival order and improve the answers delivered under communication limits. We will test when nodes receiving the same evidence converge to a consistent event picture and how accurately they answer standing questions when communication is limited.

Research plan.

- **Extract updateable event facts, not overwritten summaries.** Turn messages into small event updates recording actors, places, times, relationships, corrections, and the underlying source. Maintain explicit alternatives when accounts conflict. Event identities and links will be revised conservatively as additional evidence arrives, so a late message does not silently erase a newer correction.
- **Fuse reports independently of arrival order.** Design associative, commutative, and idempotent update rules for the symbolic event records, while studying the harder cases introduced by uncertain entity coreference and LLM extraction. We will measure when nodes receiving identical evidence converge and identify the conditions in which extraction ambiguity prevents convergence.
- **Share information by its effect on live questions.** Express critical information requirements as standing queries over event knowledge. Under a limited channel, prioritize updates that change the answer, remove a high-impact contradiction, or narrow uncertainty. Each node will answer with traceable supporting reports and an indication of what it has not yet received.

Evaluation and expected outcomes. Multi-document event corpora built from KAIROS-style scenarios and MAVEN-ERE, plus MEQA questions, will be partitioned across simulated nodes with message delay, duplication, loss, and reordering. We will compare graph agreement, query accuracy, delivery latency, and bandwidth against centralized extraction and last-writer-wins fusion.

Naval relevance. During distributed maritime operations, afloat and shore nodes may receive an unidentified-vessel report and its later identification in different orders. Updating the shared event picture using small, question-relevant messages would support a common tactical picture even across intermittent links. The framing aligns with the Command and Control portfolio's interests in uncertain, out-of-sequence information and automated integration across information stores.

Prior work and PI qualifications. Our QA-based event extraction (Du and Cardie, 2020) and MEQA benchmark (Li et al., 2024) support the text-to-event foundation. The PI holds a 2024 NSF CAREER award on consistent event graphs. As a postdoc, the PI led system integration for the RESIN-11 team in DARPA KAIROS.